

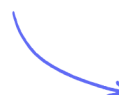
Multiple Choice Questions

Ionic & Metallic Bonding & Structure

Evidence of Ions / Ionic Bonding & Structures / Ionic Bond Strength / Polarisation /
Metallic Bonding & Lattices

Easy (6 questions)	/7
Medium (7 questions)	/7
Total Marks	/14

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Easy Questions

- 1 (a) Element **X** is in Group 2 of the Periodic Table and element **Y** is in Group 7. **X** and **Y** are not the symbols of the elements.

What is the formula of the compound formed from **X** and **Y**?

- A. XY
- B. X_2Y
- C. XY_2
- D. X_2Y_2

(1 mark)

- (b) Under what conditions does the compound formed from **X** and **Y** conduct electricity?

- A. in the solid state and in the liquid state and in aqueous solution
- B. in the solid state and in aqueous solution only
- C. in the solid state and in the liquid state only
- D. in the liquid state and in aqueous solution only

(1 mark)

- 2 Which of these compounds would you expect to have the highest melting temperature?

- A. NaCl
- B. NaF
- C. KCl
- D. KF

(1 mark)

- 3 A drop of an aqueous solution of green copper(II) chromate(VI) is placed in the centre of a strip of damp filter paper.

The ends of the filter paper are connected to a DC power supply.

What is observed after a few minutes?

- A. a green colour has moved to the negative end
- B. a green colour has moved to the positive end
- C. a yellow colour has moved to the positive end and a blue colour to the negative end
- D. a blue colour has moved to the positive end and a yellow colour to the negative end

(1 mark)

- 4 Which of these isoelectronic ions has the smallest ionic radius?

- A. N^{3-}
- B. F^-
- C. Na^+
- D. Al^{3+}

(1 mark)

- 5 Which of these has the **greatest** electrical conductivity?

- A. $\text{SF}_6(\text{g})$
- B. $\text{H}_2\text{O}(\text{l})$
- C. $\text{Hg}(\text{l})$
- D. $\text{Na}_2\text{O}(\text{s})$

(1 mark)

6 Which of these does **not** have a structure formed by a giant lattice of carbon atoms?

A. C₆₀ fullerene

B. diamond

C. graphene

D. graphite

(1 mark)

Medium Questions

1 Which properties of a **cation** result in the greatest polarising power?

- A. large radius and large charge
- B. large radius and small charge
- C. small radius and small charge
- D. small radius and large charge

(1 mark)

2 In which series are the ions in order of **decreasing** ionic radius?

- A. $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$
- B. $\text{Li}^+ > \text{Na}^+ > \text{K}^+$
- C. $\text{N}^{3-} > \text{O}^{2-} > \text{F}^-$
- D. $\text{O}^{2-} > \text{S}^{2-} > \text{Se}^{2-}$

(1 mark)

3 Which properties of an **anion** result in it being most easily polarised?

- A. large radius and large charge
- B. large radius and small charge
- C. small radius and small charge
- D. small radius and large charge

(1 mark)

4 Which of these isoelectronic ions has the **largest** radius?

- A. Na^+
- B. Mg^{2+}
- C. O^{2-}
- D. F^-

(1 mark)

5 Which ion is the most polarisable?



(1 mark)

6 Which of these ions has the **greatest** ionic radius?



(1 mark)

7 Which of these ions has the **greatest** polarising power?



(1 mark)